

Modern HTML & CSS Cheat Sheet – Portfolio Project

1 HTML5 Semantic Tags | Tag | Purpose | Mini Example | Visual | |-----|-----|-----|-----| |
<header> | Header of page or section (logo + nav) | <header><h1>Logo</h1></header> | 🌀 Top bar
with site name & menu | | <nav> | Navigation menu | <nav>Home</nav> | 🟢 Menu links | | <section> | Logical sections (hero, about, skills) | <section
id="about">...</section> | 🌟 Separate content blocks | | <footer> | Page footer |
<footer>© 2025 Suresh Nepali</footer> | ⬅️ Bottom bar with copyright | | | Inline text
element (styling/animation) | <h3>I'm a </h3> | # Inline
animated text | | + loading="eager" | Embed images & prioritize loading | | 🟡 Image shown immediately | | <a> + target="_blank" | Links, open in new tab |
GitHub | ✅ Opens new browser tab |

2 CSS Variables & Root

```
:root {  
  --bg: #f5f3ef;  
  --primary: #4a3f35;  
  --accent: #c19a6b;  
}
```

- :root → top-level HTML element - --variable → defines reusable values - var(--variable) →
uses the variable anywhere

Example Usage:

```
body {  
  background-color: var(--bg);  
  color: var(--primary);  
}
```

Visual: ⬅️ Centralized colors make styling consistent and easy to manage.

3 Flexbox

```
.head {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
}
```

- `display: flex;` → makes container flexible - `justify-content` → horizontal alignment - `align-items` → vertical alignment - `flex-wrap` → wraps items to the next line

Visual: [Item 1] [Item 2] [Item 3]

Example HTML:

```
<div class="head">
  <h1>Logo</h1>
  <nav>Menu</nav>
</div>
```

4 CSS Grid

```
.work-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 20px;
}
```

- `display: grid;` → enables grid layout - `grid-template-columns` → defines columns - `repeat(auto-fit, minmax(...))` → responsive columns - `gap` → space between grid items

Visual: [Card] [Card] [Card] [Card] [Card] [Card]

Mini Example:

```
<div class="skill-grid">
  <div>HTML5</div>
  <div>CSS3</div>
  <div>JS</div>
</div>
```

5 Aspect Ratio

```
.certificate-card {
  aspect-ratio: 4 / 3;
}
```

- Ensures consistent width:height ratio - Auto-resizes for responsiveness Visual: +-----+ | | | 4:3 ratio
+-----+

6 Object-Fit

```
img {  
  object-fit: cover;  
}
```

- Makes image fill container without distortion - Options: cover, contain, fill Visual: [Image fits nicely inside box without stretching]

7 Transitions & Hover Effects

```
.skill:hover {  
  background-color: var(--primary);  
  color: var(--bg);  
  transition: 0.3s;  
}
```

- `transition` → smooth change on hover - `hover` → triggers effect when mouse is over element Visual: Normal: [HTML5] → Hover: [HTML5 changes color smoothly]

8 Media Queries (Responsive Design)

```
@media (max-width: 600px) {  
  .contact h2 {  
    font-size: 1.7rem;  
  }  
}
```

- Makes layout responsive for mobile - Applies CSS only if screen width $\leq 600\text{px}$ Visual: Desktop → large header, Mobile → smaller header

9 JavaScript Animation (Dynamic Text)

```
const roles = ["Coder", "Programmer", "Designer"];  
let i = 0;  
const roleSpan = document.querySelector(".animated-role");  
  
function changeRole() {  
  roleSpan.textContent = roles[i];  
  i = (i + 1) % roles.length;  
}
```

```
setInterval(changeRole, 2000);
```

- Cycles through roles every 2 seconds - Updates `<span>` text dynamically Visual: Hello, I'm a Coder → Programmer → Designer → Coder...

10 Box Shadow & Border Radius

```
.work-card {  
  border-radius: 15px;  
  box-shadow: 0 4px 15px rgba(0,0,0,0.1);  
}
```

- `border-radius` → rounded corners - `box-shadow` → soft shadow for depth Visual: +-----+ | Card Box | +-----+

1 **1** Cheat Sheet Summary Table | Feature | CSS/HTML | Use | Visual | |-----|-----|----|-----| | Semantic Tags | `<header>`, `<section>`, `<footer>` | Structure & SEO |  Blocks | | CSS Variables | `--primary` | Reusable colors |  Palette | | Flexbox | `display: flex;` | Horizontal/Vertical alignment |  | | CSS Grid | `display: grid;` | Layout cards & items |  | | Aspect Ratio | `aspect-ratio: 4/3;` | Fixed proportions |  | | Object-Fit | `object-fit: cover;` | Images inside containers |  有 | | Transition | `transition: 0.3s;` | Smooth hover effect | `#` | | Media Queries | `@media` | Responsive design |  | | JS Dynamic Text | `setInterval()` | Animated roles |  | | Box Shadow | `box-shadow` | Depth |  | | Border Radius | `border-radius` | Rounded corners |  |